Equations

Equations show the _____ and products of a chemical reaction.

The reactants=_____
_____
__.

Products=_____
_____
___

This can be done as a word or formula equation.

When writing a formula equations from a word equations there are certain formula that need to be remembered. Use the information from last lesson to complete the following.

Diatomic gasses. These are gasses which have molecules made of two atoms of the same elements.

Gas name	Formula
Oxygen	
Nitrogen	
Chlorine	
Hydrogen	

There are common formula that need to be remembered.

Compound name	Formula
Water	
Carbon dioxide	
Methane	CH_4
Elemental carbon	C

There are also common acids that have set formula

Acid name	Formula
Nitric acid	HNO_3
Sulphuric acid	H_2SO_4
Phosphoric acid	H_3PO_4
Carbonic acid	H_2CO_3

For metals in a chemical reaction just the symbol of the metal is written.

For ionic compounds the formula is worked out using the “drop and swap” method and the table of common ions. This table is given at the end of this booklet.

Use this information to convert the word equations on the next pages into formula equations. The first one has been done for you.

Lead metal + Nitric acid ⟶ Lead nitrate + Hydrogen gas

$Pb + HNO_3 \longrightarrow Pb(NO_3)_2 + H_2$

There is no need to balance any equations at this stage.

Write the formula equations under the following word equations.

Methane gas + oxygen gas ⟶ Carbon dioxide + water

Hydrogen gas + oxygen gas ⟶ water

Sodium metal + Hydrochloric acid ⟶ sodium chloride + hydrogen gas

Lead oxide + elemental carbon ⟶ Lead metal + carbon dioxide

Silver metal + hydrochloric acid ⟶ silver chloride + Hydrogen gas

Iron metal + oxygen gas ⟶ Iron oxide

Iron oxide + Hydrogen ⟶ iron metal + water

Potassium metal + Chlorine gas → potassium bromide

Sodium metal + water → Sodium hydroxide + hydrogen gas

Elemental carbon + hydrogen gas → methane gas

Aluminium metal + iron oxide → aluminium oxide + iron metal

Iron metal + hydrochloric acid → iron chloride + Hydrogen gas

zinc metal + oxygen gas → zinc oxide

Z D C L A T E M L A D G E D C
G G R I M Z U P L R A Z L E O
Z P R O R D P U O S Q G E C M
I T G E P U M W J K P N M N M
F Z G B A R F P H R M K E A O
H Y D R O C H L O R I C N L N
T J R F O J T D U R P C T A P
Q R J K R P U A M S E R A B E
F K C S R C H A N E K T L E C
C I M O T A I D A T T F A A S
W A F S Y M B O L S S H R W W
O Q Q G D F A W H E H B A X A
R B O K F L Q N F U O X E N P
R Y N E C G Q Y T N K X D R E
A I O N S V S T N Z B J A X P

ARROW
BALANCED
CARBON
COMMON
DIATOMIC
DROP
ELEMENTAL
FORMULA
GAS
HYDROCHLORIC
IONS
METAL
METHANE
PRODUCTS
REACTANTS
SULFURIC
SWAP
SYMBOLS
WATER
WORD

Positive Ions (cations)		Negative Ions (anions)	
Name	Formula	Name	Formula
Hydrogen	H^+	Chloride	Cl^-
Sodium	Na^+	Bromide	Br^-
Silver	Ag^+	Fluoride	F^-
Potasssium	K^+	Iodide	I^-
Lithium	Li^+	Hydroxide	OH^-
Ammonium	NH_4^+	Nitrate	NO_3^-
Barium	Ba^{2+}	Oxide	O^{2-}
Calcium	Ca^{2+}	Sulphide	S^{2-}
Copper(II)	Cu^{2+}	Sulphate	SO_4^{2-}
Magnesium	Mg^{2+}	Carbonate	CO_3^{2-}
Zinc	Zn^{2+}	Hydrogencarbonate	HCO_3^-
Lead	Pb^{2+}		
Iron(II)	Fe^{2+}		
Iron(III)	Fe^{3+}		
Aluminium	Al^{3+}		